
11.1 API Overview

Document Information

Field	Value
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1. Introduction

1.1 Purpose

This document provides a comprehensive overview of the NewPOPSys API architecture, including authentication mechanisms, versioning strategies, rate limiting policies, and error handling conventions. It serves as the foundational reference for all API consumers and integrators.

1.2 Scope

The NewPOPSys API enables:

- **Inbound Operations:** PSP systems acknowledging orders, updating statuses, and creating shipments
- **Outbound Webhooks:** Event-driven notifications to integrated systems
- **Internal Services:** Survey management, store layouts, photo rules, and health monitoring

1.3 API Version

Attribute	Value
Current Version	1.0.0
OpenAPI Specification	3.1.0
Base Path	/api/v1

2. Server Environments

2.1 Environment Configuration

Environment	Base URL	Purpose
Production	https://api.newpopsys.com/v1	Live production traffic
Staging	https://staging-api.newpopsys.com/v1	Pre-production testing

Development	http://localhost:3000/v1	Local development
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2.2 Environment Identification

API keys are prefixed to identify their environment:

Prefix	Environment	Example
vg_live_	Production	vg_live_abc123def456
vg_test_	Staging/Development	vg_test_xyz789ghi012

3. Authentication

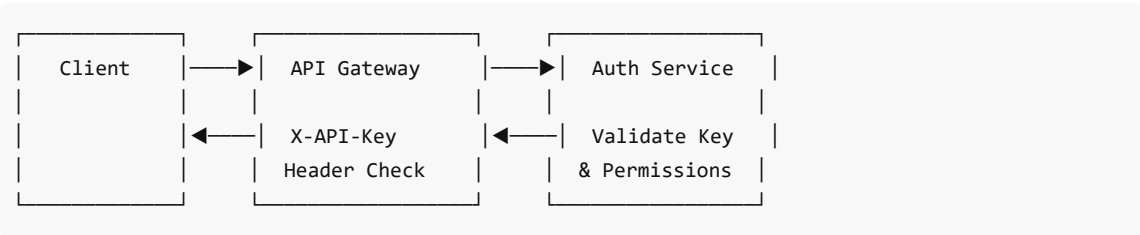
3.1 API Key Authentication

NewPOPSys uses API Key authentication for all API requests.

3.1.1 Request Header Format

X-API-Key: vg_live_your_api_key_here

3.1.2 Authentication Flow



3.2 API Key Security Requirements

Requirement	Description
Transmission	HTTPS/TLS 1.3 required
Storage	Encrypted at rest, never in logs
Rotation	Minimum every 90 days
Revocation	Immediate effect upon deletion
Scope	Tenant-specific permissions

3.3 Authentication Errors

HTTP Code	Error	Description
401	UNAUTHORIZED	Missing or invalid API key
403	FORBIDDEN	Valid key but insufficient permissions

Example Error Response:

```
{
  "error": {
    "code": "UNAUTHORIZED",
    "message": "Invalid or missing API key",
    "timestamp": "2025-01-13T10:30:00Z",
    "requestId": "req_abc123def456"
  }
}
```

4. API Versioning

4.1 Versioning Strategy

NewPOPSys employs **URL Path Versioning** for explicit version control.

Strategy	Format	Example
URL Path	/v{major}	/v1/orders

4.2 Version Lifecycle

Phase	Duration	Description
Current	Active	Full support and new features
Deprecated	6 months	Security updates only, migration recommended
Sunset	3 months	Read-only, then decommissioned

4.3 Breaking vs Non-Breaking Changes

Non-Breaking Changes (No Version Bump):

- Adding new optional request parameters
- Adding new response fields
- Adding new endpoints
- Adding new webhook event types

Breaking Changes (Major Version Bump):

- Removing or renaming fields
- Changing field types
- Modifying required parameters
- Changing endpoint paths
- Altering authentication mechanisms

4.4 Deprecation Headers

When endpoints are deprecated, responses include:

`Deprecation: true`
`Sunset: Sat, 01 Jul 2025 00:00:00 GMT`
`Link: <https://docs.newpopsys.com/migration>; rel="deprecation"`

5. Rate Limiting

5.1 Rate Limit Tiers

Category	Limit	Window	Scope
Authentication	20 requests	1 minute	Per IP
General API	300 requests	1 minute	Per API Key
File Uploads	60 requests	1 minute	Per API Key
Data Exports	10 requests	1 minute	Per Tenant
Webhooks	1000 events	1 minute	Per Endpoint

5.2 Rate Limit Response Headers

Header	Description	Example
X-RateLimit-Limit	Maximum requests allowed	300
X-RateLimit-Remaining	Requests remaining in window	247
X-RateLimit-Reset	Unix timestamp when limit resets	1705142400

Example Response Headers:

```
HTTP/1.1 200 OK
X-RateLimit-Limit: 300
X-RateLimit-Remaining: 247
X-RateLimit-Reset: 1705142400
Content-Type: application/json
```

5.3 Rate Limit Exceeded Response

When rate limits are exceeded, the API returns HTTP 429:

```
{
  "error": {
    "code": "RATE_LIMIT_EXCEEDED",
    "message": "Too many requests. Please retry after 45 seconds.",
    "retryAfter": 45,
    "timestamp": "2025-01-13T10:30:00Z",
    "requestId": "req_abc123def456"
  }
}
```

Response Headers:

HTTP/1.1 429 Too Many Requests
Retry-After: 45
X-RateLimit-Limit: 300
X-RateLimit-Remaining: 0
X-RateLimit-Reset: 1705142445

5.4 Rate Limit Best Practices

Practice	Description
Exponential Backoff	Double wait time on each 429 response
Request Batching	Combine multiple operations where possible
Caching	Cache responses to reduce API calls
Queue Management	Implement client-side request queuing

6. Request Standards

6.1 Content Types

Operation	Content-Type	Accept
JSON Requests	application/json	application/json
File Uploads	multipart/form-data	application/json
Binary Downloads	N/A	application/octet-stream

6.2 Request Headers

Header	Required	Description
X-API-Key	Yes	API authentication key
Content-Type	Yes*	Request body format (*for POST/PUT/PATCH)
Accept	No	Preferred response format
Idempotency-Key	Recommended	Prevents duplicate operations
X-Request-ID	No	Client-provided correlation ID

6.3 Idempotency

For POST, PUT, and PATCH requests, use the Idempotency-Key header to ensure exactly-once semantics:

POST /v1/orders/ORD-001/shipments HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123

Idempotency-Key: idem_7f8e9d0c-1a2b-3c4d-5e6f-7a8b9c0d1e2f
Content-Type: application/json

```
{
  "carrier": "UPS",
  "trackingNumber": "1Z999AA10123456784"
}
```

Idempotency Behavior:

Scenario	Response
First request	Normal processing, result cached for 24 hours
Duplicate key (same body)	Return cached response (HTTP 200)
Duplicate key (different body)	Return HTTP 409 Conflict

7. Response Standards

7.1 HTTP Status Codes

Success Codes

Code	Status	Usage
200	OK	Successful GET, PUT, PATCH
201	Created	Successful POST creating resource
202	Accepted	Async operation queued
204	No Content	Successful DELETE

Client Error Codes

Code	Status	Usage
400	Bad Request	Invalid request body or parameters
401	Unauthorized	Missing or invalid API key
403	Forbidden	Insufficient permissions
404	Not Found	Resource does not exist
409	Conflict	Resource state conflict
422	Unprocessable Entity	Validation failed
429	Too Many Requests	Rate limit exceeded

Server Error Codes

Code	Status	Usage
------	--------	-------

500	Internal Server Error	Unexpected server error
502	Bad Gateway	Upstream service failure
503	Service Unavailable	Temporary maintenance
504	Gateway Timeout	Upstream service timeout

7.2 Standard Response Format

Success Response

```
{
  "data": {
    "id": "ORD-2025-001234",
    "status": "acknowledged",
    "updatedAt": "2025-01-13T10:30:00Z"
  },
  "meta": {
    "requestId": "req_abc123def456",
    "timestamp": "2025-01-13T10:30:00Z"
  }
}
```

Collection Response (with Pagination)

```
{
  "data": [
    { "id": "ORD-001", "status": "pending" },
    { "id": "ORD-002", "status": "shipped" }
  ],
  "meta": {
    "requestId": "req_abc123def456",
    "timestamp": "2025-01-13T10:30:00Z"
  },
  "pagination": {
    "page": 1,
    "pageSize": 20,
    "totalItems": 156,
    "totalPages": 8,
    "hasNextPage": true,
    "hasPreviousPage": false
  }
}
```

7.3 Error Response Format

All error responses follow a consistent structure:

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "details": [
      {
        "field": "trackingNumber",
        "message": "Tracking number is required",
        "code": "REQUIRED_FIELD"
      },
      {
        "field": "carrier",
        "message": "Invalid carrier code",
        "code": "INVALID_VALUE",
        "allowedValues": ["UPS", "FEDEX", "USPS", "DHL"]
      }
    ],
    "timestamp": "2025-01-13T10:30:00Z",
    "requestId": "req_abc123def456",
    "documentationUrl": "https://docs.newpopsys.com/errors/VALIDATION_ERROR"
  }
}
```

7.4 Error Code Reference

Error Code	HTTP Status	Description
UNAUTHORIZED	401	Invalid or missing API key
FORBIDDEN	403	Insufficient permissions
NOT_FOUND	404	Resource not found
VALIDATION_ERROR	400/422	Request validation failed
CONFLICT	409	Resource state conflict
RATE_LIMIT_EXCEEDED	429	Too many requests
INTERNAL_ERROR	500	Unexpected server error
SERVICE_UNAVAILABLE	503	Service temporarily unavailable
IDEMPOTENCY_CONFLICT	409	Idempotency key reused with different payload

8. Pagination

8.1 Pagination Parameters

Parameter	Type	Default	Max	Description
page	integer	1	-	Page number (1-indexed)

pageSize	integer	20	100	Items per page
sort	string	varies	-	Sort field and direction

8.2 Example Request

```
GET /v1/orders?page=2&pageSize=50&sort=-createdAt HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
```

8.3 Sort Direction

Prefix	Direction	Example
(none)	Ascending	sort=createdAt
-	Descending	sort=-createdAt

9. Health Check Endpoints

9.1 Endpoint Summary

Endpoint	Method	Description
/health	GET	Basic health check
/health/ready	GET	Readiness probe (all dependencies)
/health/live	GET	Liveness probe (service running)

9.2 Health Response

```
{
  "status": "healthy",
  "version": "1.0.0",
  "timestamp": "2025-01-13T10:30:00Z",
  "checks": {
    "database": "healthy",
    "redis": "healthy",
    "queue": "healthy"
  }
}
```

9.3 Health Status Values

Status	Description
healthy	All systems operational
degraded	Some non-critical services impaired

unhealthy	Critical services unavailable
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10. Security Considerations

10.1 Transport Security

Requirement	Specification
Protocol	HTTPS only (HTTP redirected)
TLS Version	1.2 minimum, 1.3 preferred
Certificate	Valid, non-expired, trusted CA
HSTS	Enabled with 1-year max-age

10.2 Request Validation

Check	Description
Input Sanitization	All inputs validated and sanitized
Size Limits	Request body max 10MB
Content-Type	Strict matching required
SQL Injection	Parameterized queries only
XSS Prevention	Output encoding applied

10.3 Audit Logging

All API requests are logged with:

- Request timestamp
- API key (masked)
- Request ID
- Endpoint and method
- Response status code
- Response time
- Client IP (anonymized for GDPR)

11. SDK and Client Libraries

11.1 Official SDKs

Language	Package	Repository
JavaScript/TypeScript	@newpopsys/sdk	GitHub
Python	newpopsys-python	PyPI
C#	NewPOPSys.SDK	NuGet

11.2 API Client Generation

OpenAPI specification available for generating custom clients:

```
https://api.newpopsys.com/v1/openapi.json
https://api.newpopsys.com/v1/openapi.yaml
```

12. Related Documents

Document ID	Title	Description
SRS-11.2	Internal APIs	Detailed endpoint specifications
SRS-11.3	External Integrations	Third-party integration details
SRS-11.4	Webhooks	Event-driven notification system
SRS-3.4	Integration Architecture	Technical integration patterns

Revision History

Version	Date	Author	Changes
1.0	2025-01-13	System	Initial version

11.2 Internal APIs

Document Information

Field	Value
Document ID	SRS-11.2
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Parent Document	SRS-11 API Specifications

1. Introduction

1.1 Purpose

This document specifies all internal RESTful API endpoints for the NewPOPSys platform. These APIs enable PSP (Print Service Provider) systems and internal services to interact with the NewPOPSys platform for order management, shipment tracking, and related operations.

1.2 Base URL

All endpoints are relative to the base URL:

- **Production:** https://api.newpopsys.com/v1
- **Staging:** https://staging-api.newpopsys.com/v1
- **Development:** http://localhost:3000/v1

1.3 Authentication

All endpoints require API key authentication via the X-API-Key header unless otherwise specified.

2. Orders API

2.1 Endpoint Summary

Method	Endpoint	Description
POST	/orders/{orderId}/acknowledge	Acknowledge order receipt
PUT	/orders/{orderId}/status	Update order status
POST	/orders/{orderId}/shipments	Create shipment for order

2.2 Acknowledge Order

Acknowledges receipt of an order from NewPOPSys, confirming the PSP has received and will process the order.

Request

```
POST /v1/orders/{orderId}/acknowledge HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

{
  "acknowledgedAt": "2025-01-13T10:30:00Z",
  "estimatedShipDate": "2025-01-15",
  "notes": "Order received and queued for production"
}
```

Path Parameters

Parameter	Type	Required	Description
orderId	string	Yes	Unique order identifier (e.g., ORD-2025-001234)

Request Body

Field	Type	Required	Description
acknowledgedAt	string (ISO 8601)	Yes	Timestamp of acknowledgment

estimatedShipDate	string (date)	No	Expected ship date (YYYY-MM-DD)
notes	string	No	Optional notes (max 500 chars)

Response

Success (200 OK)

```
{
  "data": {
    "orderId": "ORD-2025-001234",
    "status": "acknowledged",
    "acknowledgedAt": "2025-01-13T10:30:00Z",
    "estimatedShipDate": "2025-01-15"
  },
  "meta": {
    "requestId": "req_abc123def456",
    "timestamp": "2025-01-13T10:30:00Z"
  }
}
```

Error Responses

Code	Error	Description
400	VALIDATION_ERROR	Invalid request body
401	UNAUTHORIZED	Invalid API key
404	ORDER_NOT_FOUND	Order does not exist
409	ALREADY_ACKNOWLEDGED	Order already acknowledged

2.3 Update Order Status

Updates the current status of an order in the fulfillment workflow.

Request

```
PUT /v1/orders/{orderId}/status HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

{
  "status": "in_production",
  "statusDetails": "Printing in progress - 50% complete",
  "updatedAt": "2025-01-13T14:30:00Z"
}
```

Path Parameters

Parameter	Type	Required	Description
orderId	string	Yes	Unique order identifier

Request Body

Field	Type	Required	Description
status	string	Yes	New order status
statusDetails	string	No	Additional status information
updatedAt	string (ISO 8601)	Yes	Timestamp of status change

Valid Status Values

Status	Description
acknowledged	Order received by PSP
in_production	Order being manufactured
quality_check	Quality inspection in progress
ready_to_ship	Production complete, awaiting shipment
shipped	Order shipped (use shipment endpoint)
on_hold	Order temporarily paused
cancelled	Order cancelled

Response

Success (200 OK)

```
{
  "data": {
    "orderId": "ORD-2025-001234",
    "previousStatus": "acknowledged",
    "currentStatus": "in_production",
    "statusDetails": "Printing in progress - 50% complete",
    "updatedAt": "2025-01-13T14:30:00Z"
  },
  "meta": {
    "requestId": "req_def456ghi789",
    "timestamp": "2025-01-13T14:30:00Z"
  }
}
```

Error Responses

Code	Error	Description
------	-------	-------------

400	INVALID_STATUS	Invalid status value
400	INVALID_TRANSITION	Status transition not allowed
404	ORDER_NOT_FOUND	Order does not exist

3. Shipments API

3.1 Endpoint Summary

Method	Endpoint	Description
POST	/orders/{orderId}/shipments	Create new shipment
PUT	/shipments/{shipmentId}/delivered	Confirm delivery

3.2 Create Shipment

Creates a shipment record for an order with tracking information.

Request

```
POST /v1/orders/{orderId}/shipments HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json
Idempotency-Key: idem_7f8e9d0c-1a2b-3c4d-5e6f-7a8b9c0d1e2f

{
  "carrier": "UPS",
  "trackingNumber": "1Z999AA10123456784",
  "serviceType": "GROUND",
  "shippedAt": "2025-01-15T09:00:00Z",
  "estimatedDelivery": "2025-01-18",
  "packages": [
    {
      "weight": 2.5,
      "weightUnit": "lb",
      "dimensions": {
        "length": 12,
        "width": 10,
        "height": 4,
        "unit": "in"
      },
    },
    "contents": "POP Display Kit - Store #1234"
  ]
},
"shipFrom": {
  "name": "PSP Fulfillment Center",
  "address1": "100 Industrial Way",
  "city": "Chicago",
```

```
    "state": "IL",
    "postalCode": "60601",
    "country": "US"
  },
  "shipTo": {
    "name": "Store Manager",
    "company": "Retail Store #1234",
    "address1": "500 Main Street",
    "city": "Milwaukee",
    "state": "WI",
    "postalCode": "53202",
    "country": "US"
  }
}
```

Path Parameters

Parameter	Type	Required	Description
orderId	string	Yes	Order identifier for shipment

Request Body

Field	Type	Required	Description
carrier	string	Yes	Shipping carrier code
trackingNumber	string	Yes	Carrier tracking number
serviceType	string	No	Carrier service level
shippedAt	string (ISO 8601)	Yes	Shipment timestamp
estimatedDelivery	string (date)	No	Expected delivery date
packages	array	No	Package details
shipFrom	object	No	Origin address
shipTo	object	Yes	Destination address

Valid Carrier Codes

Code	Carrier Name
UPS	United Parcel Service
FEDEX	FedEx
USPS	US Postal Service
DHL	DHL Express

Response

Success (201 Created)


```
{
  "data": {
    "shipmentId": "SHP-2025-001234",
    "orderId": "ORD-2025-001234",
    "carrier": "UPS",
    "trackingNumber": "1Z999AA10123456784",
    "trackingUrl": "https://www.ups.com/track?tracknum=1Z999AA10123456784",
    "status": "in_transit",
    "shippedAt": "2025-01-15T09:00:00Z",
    "estimatedDelivery": "2025-01-18"
  },
  "meta": {
    "requestId": "req_ghi789jkl012",
    "timestamp": "2025-01-15T09:00:00Z"
  }
}
```

Error Responses

Code	Error	Description
400	INVALID_CARRIER	Unsupported carrier code
400	INVALID_TRACKING	Invalid tracking number format
404	ORDER_NOT_FOUND	Order does not exist
409	DUPLICATE_TRACKING	Tracking number already used

3.3 Confirm Delivery

Confirms that a shipment has been delivered to the destination.

Request

```
PUT /v1/shipments/{shipmentId}/delivered HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

{
  "deliveredAt": "2025-01-18T14:30:00Z",
  "signedBy": "J. Smith",
  "deliveryNotes": "Left at front desk",
  "proofOfDelivery": {
    "type": "signature",
    "imageUrl": "https://storage.example.com/pod/SHP-2025-001234.png"
  }
}
```

Path Parameters

Parameter	Type	Required	Description
shipmentId	string	Yes	Shipment identifier

Request Body

Field	Type	Required	Description
deliveredAt	string (ISO 8601)	Yes	Delivery timestamp
signedBy	string	No	Name of person who signed
deliveryNotes	string	No	Delivery notes
proofOfDelivery	object	No	Proof of delivery details

Response

Success (200 OK)

```
{
  "data": {
    "shipmentId": "SHP-2025-001234",
    "orderId": "ORD-2025-001234",
    "status": "delivered",
    "deliveredAt": "2025-01-18T14:30:00Z",
    "signedBy": "J. Smith"
  },
  "meta": {
    "requestId": "req_mno345pqr678",
    "timestamp": "2025-01-18T14:30:00Z"
  }
}
```

4. Surveys API

4.1 Endpoint Summary

Method	Endpoint	Description
GET	/surveys/templates	List survey templates
GET	/surveys/templates/{templateId}	Get template details
POST	/surveys/responses	Submit survey response
GET	/surveys/responses/{responseId}	Get response details

4.2 List Survey Templates

Retrieves available survey templates for a tenant.

Request

GET /v1/surveys/templates?page=1&pageSize=20 HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123

Query Parameters

Parameter	Type	Default	Description
page	integer	1	Page number
pageSize	integer	20	Items per page (max 100)
status	string	-	Filter by status (active, archived)
type	string	-	Filter by survey type

Response

Success (200 OK)

```
{
  "data": [
    {
      "templateId": "TMPL-001",
      "name": "Store Installation Survey",
      "description": "Post-installation verification survey",
      "version": "2.0",
      "status": "active",
      "questionCount": 15,
      "estimatedMinutes": 10,
      "createdAt": "2024-06-01T00:00:00Z",
      "updatedAt": "2024-12-15T00:00:00Z"
    },
    {
      "templateId": "TMPL-002",
      "name": "Pre-Visit Store Assessment",
      "description": "Initial store layout assessment",
      "version": "1.5",
      "status": "active",
      "questionCount": 25,
      "estimatedMinutes": 20,
      "createdAt": "2024-03-15T00:00:00Z",
      "updatedAt": "2024-11-01T00:00:00Z"
    }
  ],
  "meta": {
    "requestId": "req_stu901vwx234",
    "timestamp": "2025-01-13T10:30:00Z"
  },
  "pagination": {
```

```
"page": 1,
"pageSize": 20,
"totalItems": 8,
"totalPages": 1,
"hasNextPage": false,
"hasPreviousPage": false
}
}
```

4.3 Get Survey Template

Retrieves detailed survey template including all questions.

Request

```
GET /v1/surveys/templates/TMPL-001 HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
```

Response

Success (200 OK)

```
{
  "data": {
    "templateId": "TMPL-001",
    "name": "Store Installation Survey",
    "description": "Post-installation verification survey",
    "version": "2.0",
    "status": "active",
    "sections": [
      {
        "sectionId": "SEC-001",
        "title": "Display Installation",
        "order": 1,
        "questions": [
          {
            "questionId": "Q-001",
            "type": "multiple_choice",
            "text": "Was the display installed in the correct location?",
            "required": true,
            "options": ["Yes", "No", "Partially"]
          },
          {
            "questionId": "Q-002",
            "type": "photo",
            "text": "Take a photo of the installed display",
            "required": true,
            "validation": {
              "minPhotos": 1,
              "maxPhotos": 5,
            }
          }
        ]
      }
    ]
  }
}
```

```
        "minResolution": "1920x1080"
      }
    },
    {
      "questionId": "Q-003",
      "type": "text",
      "text": "Describe any installation issues",
      "required": false,
      "validation": {
        "maxLength": 500
      }
    }
  ]
}
],
"createdAt": "2024-06-01T00:00:00Z",
"updatedAt": "2024-12-15T00:00:00Z"
},
"meta": {
  "requestId": "req_yza567bcd890",
  "timestamp": "2025-01-13T10:30:00Z"
}
}
```

4.4 Submit Survey Response

Submits a completed survey response.

Request

```
POST /v1/surveys/responses HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json
Idempotency-Key: idem_survey_response_12345
```

```
{
  "templateId": "TMPL-001",
  "storeId": "STORE-1234",
  "campaignId": "CAMP-2025-Q1",
  "submittedBy": {
    "userId": "USR-567",
    "name": "John Installer",
    "email": "john@installer.com"
  },
  "answers": [
    {
      "questionId": "Q-001",
      "value": "Yes"
    },
    {
```

```

    "questionId": "Q-002",
    "value": {
      "photos": [
        {
          "url": "https://storage.example.com/photos/photo1.jpg",
          "takenAt": "2025-01-13T10:00:00Z",
          "location": {
            "latitude": 43.0389,
            "longitude": -87.9065
          }
        }
      ]
    },
    {
      "questionId": "Q-003",
      "value": "No issues encountered"
    }
  ],
  "completedAt": "2025-01-13T10:30:00Z",
  "deviceInfo": {
    "type": "mobile",
    "os": "iOS 17.2",
    "appVersion": "2.5.0"
  }
}

```

Response

Success (201 Created)

```

{
  "data": {
    "responseId": "RESP-2025-001234",
    "templateId": "TMPL-001",
    "storeId": "STORE-1234",
    "campaignId": "CAMP-2025-Q1",
    "status": "submitted",
    "completedAt": "2025-01-13T10:30:00Z",
    "validationStatus": "pending"
  },
  "meta": {
    "requestId": "req_efg123hij456",
    "timestamp": "2025-01-13T10:30:00Z"
  }
}

```

5. Store Layouts API

5.1 Endpoint Summary

Method	Endpoint	Description
GET	/stores/{storeId}/layouts	Get store layout
PUT	/stores/{storeId}/layouts	Update store layout
GET	/stores/{storeId}/layouts/history	Get layout change history

5.2 Get Store Layout

Retrieves the current layout configuration for a store.

Request

```
GET /v1/stores/STORE-1234/layouts HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
```

Response

Success (200 OK)

```
{
  "data": {
    "storeId": "STORE-1234",
    "storeName": "Downtown Milwaukee #1234",
    "layoutVersion": "3.2",
    "lastUpdated": "2025-01-10T08:00:00Z",
    "dimensions": {
      "width": 50,
      "length": 80,
      "unit": "feet"
    },
  },
  "zones": [
    {
      "zoneId": "ZONE-A",
      "name": "Entrance Display Area",
      "type": "high_traffic",
      "position": {
        "x": 0,
        "y": 0,
        "width": 15,
        "length": 20
      },
    },
  ],
  "fixtures": [
    {
      "fixtureId": "FIX-001",
      "type": "floor_display",
      "name": "Main Promotional Display",
      "dimensions": {
        "width": 4,
```

```
        "depth": 2,
        "height": 6,
        "unit": "feet"
      },
      "position": {
        "x": 5,
        "y": 8
      }
    }
  ]
},
{
  "zoneId": "ZONE-B",
  "name": "Checkout Area",
  "type": "point_of_sale",
  "position": {
    "x": 35,
    "y": 70,
    "width": 15,
    "length": 10
  },
  "fixtures": []
}
],
"floorPlanUrl": "https://storage.example.com/layouts/STORE-1234-v3.2.pdf"
},
"meta": {
  "requestId": "req_klm789nop012",
  "timestamp": "2025-01-13T10:30:00Z"
}
}
```

5.3 Update Store Layout

Updates the store layout configuration.

Request

```
PUT /v1/stores/STORE-1234/layouts HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

{
  "zones": [
    {
      "zoneId": "ZONE-A",
      "name": "Entrance Display Area",
      "type": "high_traffic",
      "position": {
        "x": 0,
```



```

        "y": 0,
        "width": 15,
        "length": 20
    },
    "fixtures": [
        {
            "fixtureId": "FIX-001",
            "type": "floor_display",
            "name": "Main Promotional Display",
            "position": {
                "x": 5,
                "y": 8
            }
        },
        {
            "fixtureId": "FIX-002",
            "type": "endcap",
            "name": "New Seasonal Endcap",
            "position": {
                "x": 12,
                "y": 5
            }
        }
    ]
},
"changeReason": "Added new seasonal endcap display",
"effectiveDate": "2025-01-15"
}

```

Response

Success (200 OK)

```

{
  "data": {
    "storeId": "STORE-1234",
    "layoutVersion": "3.3",
    "previousVersion": "3.2",
    "lastUpdated": "2025-01-13T10:30:00Z",
    "effectiveDate": "2025-01-15",
    "changeReason": "Added new seasonal endcap display"
  },
  "meta": {
    "requestId": "req_qrs345tuv678",
    "timestamp": "2025-01-13T10:30:00Z"
  }
}

```

6. Photo Rules API

6.1 Endpoint Summary

Method	Endpoint	Description
GET	/photo-rules	List photo validation rules
GET	/photo-rules/{ruleId}	Get specific rule
POST	/photos/validate	Validate photo against rules

6.2 List Photo Rules

Retrieves all photo validation rules for the tenant.

Request

```
GET /v1/photo-rules?category=installation HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
```

Query Parameters

Parameter	Type	Description
category	string	Filter by category (installation, verification, compliance)
status	string	Filter by status (active, inactive)

Response

Success (200 OK)

```
{
  "data": [
    {
      "ruleId": "RULE-001",
      "name": "Display Installation Photo",
      "category": "installation",
      "status": "active",
      "requirements": {
        "minResolution": {
          "width": 1920,
          "height": 1080
        },
        "maxFileSize": "10MB",
        "allowedFormats": ["jpeg", "png", "heic"],
        "requireGeoLocation": true,
        "requireTimestamp": true,
        "maxAge": "24h"
      },
      "aiValidation": {
        "enabled": true,

```

```

        "checks": [
            "display_visibility",
            "proper_installation",
            "branding_compliance"
        ],
        "minimumConfidence": 0.85
    }
},
{
    "ruleId": "RULE-002",
    "name": "Store Front Photo",
    "category": "verification",
    "status": "active",
    "requirements": {
        "minResolution": {
            "width": 1280,
            "height": 720
        },
        "maxFileSize": "5MB",
        "allowedFormats": ["jpeg", "png"],
        "requireGeoLocation": true,
        "requireTimestamp": true,
        "maxAge": "1h"
    },
    "aiValidation": {
        "enabled": true,
        "checks": [
            "store_front_visible",
            "signage_readable"
        ],
        "minimumConfidence": 0.80
    }
}
],
"meta": {
    "requestId": "req_wxy901zab234",
    "timestamp": "2025-01-13T10:30:00Z"
}
}

```

6.3 Validate Photo

Validates a photo against specified rules.

Request

```

POST /v1/photos/validate HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

```

```
{
  "photoUrl": "https://storage.example.com/uploads/photo123.jpg",
  "ruleIds": ["RULE-001"],
  "metadata": {
    "storeId": "STORE-1234",
    "campaignId": "CAMP-2025-Q1",
    "takenAt": "2025-01-13T10:00:00Z",
    "location": {
      "latitude": 43.0389,
      "longitude": -87.9065,
      "accuracy": 5
    }
  }
}
```

Response

Success (200 OK)

```
{
  "data": {
    "validationId": "VAL-2025-001234",
    "status": "passed",
    "results": [
      {
        "ruleId": "RULE-001",
        "passed": true,
        "checks": {
          "resolution": {
            "passed": true,
            "actual": {
              "width": 3024,
              "height": 4032
            },
            "required": {
              "width": 1920,
              "height": 1080
            }
          },
          "fileSize": {
            "passed": true,
            "actual": "4.2MB",
            "maximum": "10MB"
          },
          "geoLocation": {
            "passed": true,
            "distanceFromStore": "15m",
            "maximumAllowed": "100m"
          },
          "timestamp": {
            "passed": true,
            "age": "30m",

```

```

        "maximumAge": "24h"
    },
    "aiValidation": {
        "passed": true,
        "confidence": 0.92,
        "minimumRequired": 0.85,
        "detections": [
            {
                "check": "display_visibility",
                "confidence": 0.95,
                "passed": true
            },
            {
                "check": "proper_installation",
                "confidence": 0.89,
                "passed": true
            },
            {
                "check": "branding_compliance",
                "confidence": 0.91,
                "passed": true
            }
        ]
    }
}
}
}
}
}
},
"meta": {
    "requestId": "req_cde567fgh890",
    "timestamp": "2025-01-13T10:30:00Z"
}
}

```

7. Webhooks Management API

7.1 Endpoint Summary

Method	Endpoint	Description
GET	/webhooks	List webhook subscriptions
POST	/webhooks	Create webhook subscription
GET	/webhooks/{webhookId}	Get webhook details
PUT	/webhooks/{webhookId}	Update webhook
DELETE	/webhooks/{webhookId}	Delete webhook
GET	/webhooks/{webhookId}/deliveries	List delivery attempts

POST	/webhooks/{webhookId}/test	Send test event
------	----------------------------	-----------------

7.2 List Webhooks

Retrieves all webhook subscriptions for the authenticated tenant.

Request

```
GET /v1/webhooks?page=1&pageSize=20 HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
```

Response

Success (200 OK)

```
{
  "data": [
    {
      "webhookId": "WH-001",
      "url": "https://psp-system.example.com/webhooks/newpopsys",
      "status": "active",
      "events": [
        "order.generated",
        "campaign.published"
      ],
      "createdAt": "2024-06-01T00:00:00Z",
      "lastDeliveryAt": "2025-01-13T09:45:00Z",
      "successRate": 99.2
    }
  ],
  "meta": {
    "requestId": "req_ijk123lmn456",
    "timestamp": "2025-01-13T10:30:00Z"
  },
  "pagination": {
    "page": 1,
    "pageSize": 20,
    "totalItems": 3,
    "totalPages": 1
  }
}
```

7.3 Create Webhook

Creates a new webhook subscription.

Request

```
POST /v1/webhooks HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_abc123
Content-Type: application/json

{
  "url": "https://psp-system.example.com/webhooks/newpopsys",
  "events": [
    "order.generated",
    "campaign.published",
    "issue.created"
  ],
  "secret": "whsec_your_secret_key_here",
  "description": "Main PSP integration webhook",
  "headers": {
    "X-Custom-Header": "custom-value"
  }
}
```

Request Body

Field	Type	Required	Description
url	string	Yes	HTTPS endpoint URL
events	array	Yes	Event types to subscribe
secret	string	Yes	HMAC signing secret (min 32 chars)
description	string	No	Webhook description
headers	object	No	Custom headers to include

Response

Success (201 Created)

```
{
  "data": {
    "webhookId": "WH-002",
    "url": "https://psp-system.example.com/webhooks/newpopsys",
    "status": "active",
    "events": [
      "order.generated",
      "campaign.published",
      "issue.created"
    ],
    "createdAt": "2025-01-13T10:30:00Z"
  },
  "meta": {
    "requestId": "req_opq789rst012",
    "timestamp": "2025-01-13T10:30:00Z"
  }
}
```

```
}  
}
```

8. Health API

8.1 Endpoint Summary

Method	Endpoint	Auth	Description
GET	/health	No	Basic health check
GET	/health/ready	No	Readiness probe
GET	/health/live	No	Liveness probe

8.2 Health Check

Returns overall system health status.

Request

```
GET /v1/health HTTP/1.1  
Host: api.newpopsys.com
```

Response

Success (200 OK)

```
{  
  "status": "healthy",  
  "version": "1.0.0",  
  "timestamp": "2025-01-13T10:30:00Z",  
  "uptime": "45d 12h 30m",  
  "checks": {  
    "database": {  
      "status": "healthy",  
      "latency": "2ms"  
    },  
    "redis": {  
      "status": "healthy",  
      "latency": "1ms"  
    },  
    "queue": {  
      "status": "healthy",  
      "pendingJobs": 42  
    },  
    "storage": {  
      "status": "healthy",  
      "available": "85%"  
    }  
  }  
}
```



```
}  
}
```

Degraded (200 OK)

```
{  
  "status": "degraded",  
  "version": "1.0.0",  
  "timestamp": "2025-01-13T10:30:00Z",  
  "checks": {  
    "database": {  
      "status": "healthy",  
      "latency": "2ms"  
    },  
    "redis": {  
      "status": "degraded",  
      "latency": "150ms",  
      "message": "High latency detected"  
    },  
    "queue": {  
      "status": "healthy",  
      "pendingJobs": 42  
    }  
  }  
}
```

9. Common Data Types

9.1 Address Object

```
{  
  "name": "string",  
  "company": "string",  
  "address1": "string",  
  "address2": "string",  
  "city": "string",  
  "state": "string",  
  "postalCode": "string",  
  "country": "string (ISO 3166-1 alpha-2)",  
  "phone": "string",  
  "email": "string"  
}
```

9.2 GeoLocation Object

```
{  
  "latitude": "number (-90 to 90)",  
  "longitude": "number (-180 to 180)",  
}
```

```
"accuracy": "number (meters)",
"altitude": "number (meters, optional)"
}
```

9.3 Pagination Object

```
{
  "page": "integer (current page)",
  "pageSize": "integer (items per page)",
  "totalItems": "integer (total count)",
  "totalPages": "integer (total pages)",
  "hasNextPage": "boolean",
  "hasPreviousPage": "boolean"
}
```

10. Related Documents

Document ID	Title	Description
SRS-11.1	API Overview	Authentication, versioning, rate limiting
SRS-11.3	External Integrations	Third-party system integrations
SRS-11.4	Webhooks	Event-driven notifications
SRS-3.4	Integration Architecture	Technical integration patterns

Revision History

Version	Date	Author	Changes
1.0	2025-01-13	System	Initial version

11.3 External Integrations

Document Information

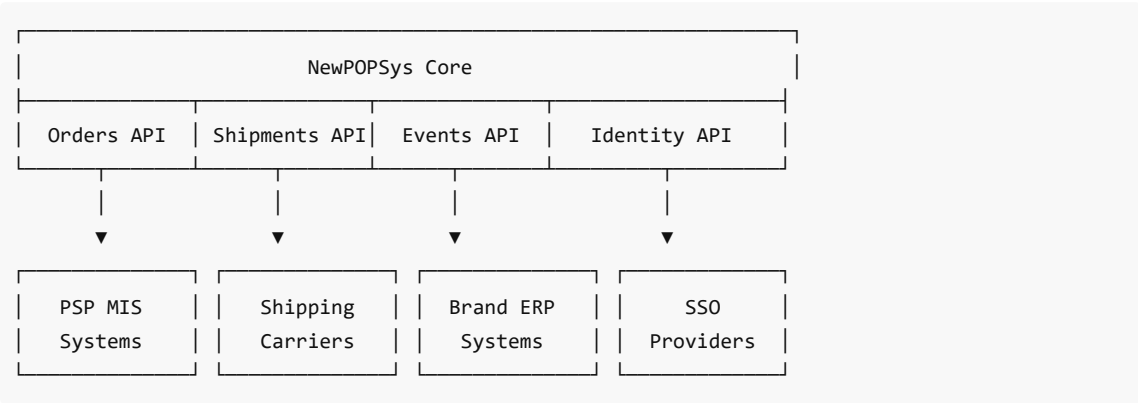
Field	Value
Document ID	SRS-11.3
Version	1.0
Last Updated	2025-01-13
Status	Draft
Parent Document	SRS-11 API Specifications

1. Introduction

1.1 Purpose

This document specifies the external system integrations for NewPOPSys, detailing how the platform connects with third-party services including ERP systems, CRM platforms, shipping carriers, and identity providers.

1.2 Integration Architecture Overview



1.3 Integration Partners Summary

Partner Category	Systems	Direction
PSP MIS	W2P, EFI, Tharstern	Bidirectional
Shipping Carriers	UPS, FedEx, USPS, DHL	Bidirectional
Brand ERP	SAP, Oracle, NetSuite	Outbound Webhooks
Identity (SSO)	Azure AD, Okta, Auth0	Inbound
Payment	Stripe, PayPal	Bidirectional
Storage	AWS S3, Azure Blob	Outbound
Notifications	SendGrid, Twilio	Outbound

2. PSP MIS Integration

2.1 Overview

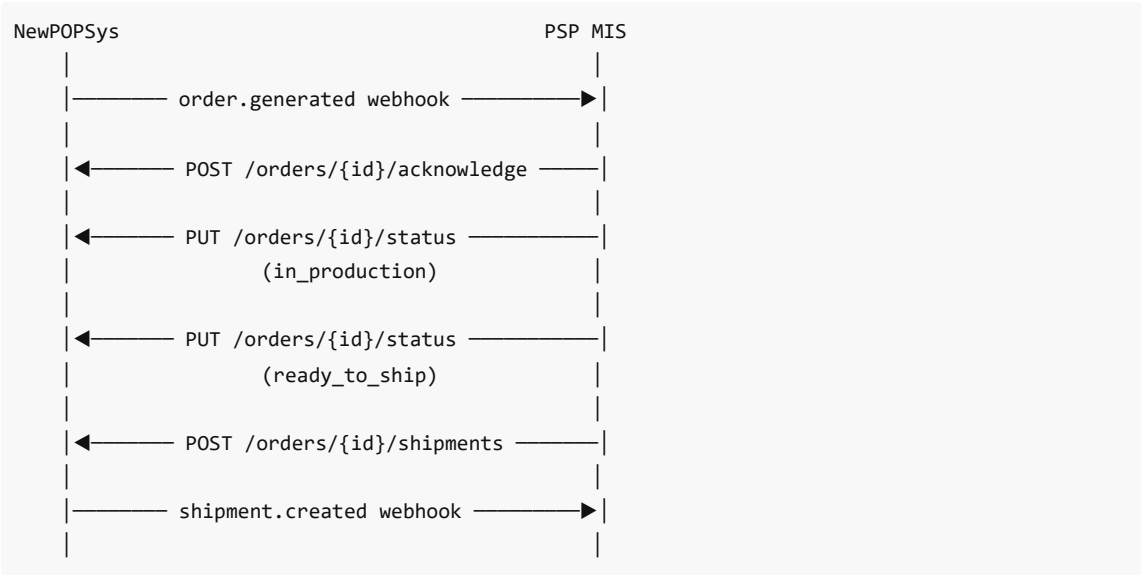
Print Service Provider (PSP) Management Information Systems receive orders from NewPOPSys and report back production status and shipment information.

2.2 Supported MIS Platforms

Platform	Version	Protocol	Authentication
W2P Custom	Various	REST API	API Key
EFI Pace	28+	REST/SOAP	OAuth 2.0

Tharstern	8+	REST API	API Key
PrintSmith	Vision	REST API	API Key

2.3 Data Flow



2.4 Order Data Mapping

2.4.1 NewPOPSys to PSP Order Fields

NewPOPSys Field	PSP MIS Field	Description
orderId	external_order_id	Unique order reference
campaignId	project_code	Campaign/project identifier
items[].sku	product_sku	Product SKU
items[].quantity	quantity	Order quantity
items[].specifications	job_specs	Print specifications
shippingAddress	delivery_address	Ship-to address
priority	rush_flag	Priority indicator
dueDate	required_date	Required delivery date

2.4.2 Sample Order Payload

```
{
  "orderId": "ORD-2025-001234",
  "campaignId": "CAMP-2025-Q1-PROMO",
  "orderDate": "2025-01-13T10:00:00Z",
  "priority": "standard",
  "dueDate": "2025-01-20",
```

```

"customer": {
  "customerId": "CUST-001",
  "name": "Acme Retail Brand",
  "accountNumber": "ACM-12345"
},
"items": [
  {
    "lineId": "LN-001",
    "sku": "POP-FLRDSP-24X36",
    "name": "Floor Display Stand 24x36",
    "quantity": 150,
    "specifications": {
      "material": "Corrugated E-Flute",
      "printing": "4/0 CMYK + Spot PMS 185C",
      "finishing": "Gloss Lamination",
      "assembly": "Ship Flat"
    },
    "artworkUrl": "https://storage.newpopsys.com/artwork/ORD-001234/LN-001.pdf"
  }
],
"shippingAddress": {
  "name": "Store Manager",
  "company": "Acme Store #1234",
  "address1": "500 Main Street",
  "city": "Milwaukee",
  "state": "WI",
  "postalCode": "53202",
  "country": "US"
},
"shippingMethod": {
  "carrier": "UPS",
  "service": "GROUND",
  "instructions": "Deliver to receiving dock"
}
}

```

2.5 PSP API Callbacks

2.5.1 Status Update Callback

```

PUT /v1/orders/ORD-2025-001234/status HTTP/1.1
Host: api.newpopsys.com
X-API-Key: vg_live_psp_key_abc123
Content-Type: application/json

{
  "status": "in_production",
  "statusDetails": "Printing complete, moving to finishing",
  "updatedAt": "2025-01-14T14:30:00Z",
  "completionPercentage": 60,

```

```
"estimatedCompletion": "2025-01-15T12:00:00Z"
}
```

2.6 Error Handling

PSP Error	NewPOPSys Handling
Order not found	Return 404, notify operations
Invalid status transition	Return 400, log attempt
Duplicate acknowledgment	Return 409, idempotent
Production failure	Accept callback, trigger issue.created

3. Shipping Carrier Integration

3.1 Supported Carriers

Carrier	API Version	Services Supported
UPS	REST 2.0	Ground, 2-Day, Next Day, Freight
FedEx	REST 2023	Ground, Express, Freight
USPS	Web Tools 3.0	Priority, Priority Express, Parcel Select
DHL	Express API 2.0	Express, eCommerce

3.2 Carrier API Endpoints

3.2.1 UPS Integration

Operation	Endpoint	Method
Rate Quote	/rating/v1/rate	POST
Create Shipment	/shipments/v1/shipments	POST
Track Package	/track/v1/details/{trackingNumber}	GET
Void Shipment	/shipments/v1/shipments/{shipmentId}/void	PUT

3.2.2 FedEx Integration

Operation	Endpoint	Method
Rate Quote	/rate/v1/rates/quotes	POST
Create Shipment	/ship/v1/shipments	POST
Track Package	/track/v1/trackingnumbers	POST
Cancel Shipment	/ship/v1/shipments/cancel	PUT

3.3 Tracking Integration

3.3.1 Polling Configuration

Carrier	Poll Interval	Batch Size	Timeout
UPS	30 minutes	100 tracking numbers	30 seconds
FedEx	30 minutes	100 tracking numbers	30 seconds
USPS	60 minutes	50 tracking numbers	60 seconds
DHL	30 minutes	50 tracking numbers	45 seconds

3.3.2 Tracking Event Mapping

Carrier Event	NewPOPSys Status	Description
Picked Up	in_transit	Package collected
In Transit	in_transit	En route
Out for Delivery	out_for_delivery	Final mile
Delivered	delivered	Package delivered
Exception	exception	Delivery issue
Returned	returned	Return to sender

3.3.3 Sample Tracking Response Mapping

UPS Tracking Response:

```
{
  "trackResponse": {
    "shipment": [
      {
        "inquiryNumber": "1Z999AA10123456784",
        "package": [
          {
            "trackingNumber": "1Z999AA10123456784",
            "activity": [
              {
                "status": {
                  "type": "D",
                  "description": "Delivered",
                  "code": "KB"
                },
                "date": "20250118",
                "time": "143000",
                "location": {
                  "city": "MILWAUKEE",
                  "stateProvince": "WI",
                  "postalCode": "53202",

```

```
        "country": "US"
      }
    }
  ]
}
}
```

Mapped NewPOPSys Event:

```
{
  "shipmentId": "SHP-2025-001234",
  "trackingNumber": "1Z999AA10123456784",
  "carrier": "UPS",
  "status": "delivered",
  "eventTimestamp": "2025-01-18T14:30:00Z",
  "location": {
    "city": "Milwaukee",
    "state": "WI",
    "postalCode": "53202",
    "country": "US"
  },
  "signedBy": "J SMITH"
}
```

3.4 Label Generation

3.4.1 Label Format Support

Carrier	Formats	Default
UPS	ZPL, PNG, PDF, GIF	PDF
FedEx	ZPL, PNG, PDF	PDF
USPS	ZPL, PDF	PDF
DHL	ZPL, PDF	PDF

3.4.2 Label Request Example

```
{
  "carrier": "UPS",
  "service": "GROUND",
  "labelFormat": "PDF",
  "shipFrom": {
    "name": "PSP Fulfillment",
    "address1": "100 Industrial Way",
    "city": "Chicago",
```



```
    "state": "IL",
    "postalCode": "60601",
    "country": "US",
    "phone": "3125551234"
  },
  "shipTo": {
    "name": "Store Manager",
    "company": "Retail Store #1234",
    "address1": "500 Main Street",
    "city": "Milwaukee",
    "state": "WI",
    "postalCode": "53202",
    "country": "US",
    "phone": "4145556789"
  },
  "packages": [
    {
      "weight": {
        "value": 5.5,
        "unit": "LB"
      },
      "dimensions": {
        "length": 12,
        "width": 10,
        "height": 4,
        "unit": "IN"
      },
      "reference1": "ORD-2025-001234",
      "reference2": "STORE-1234"
    }
  ]
}
```

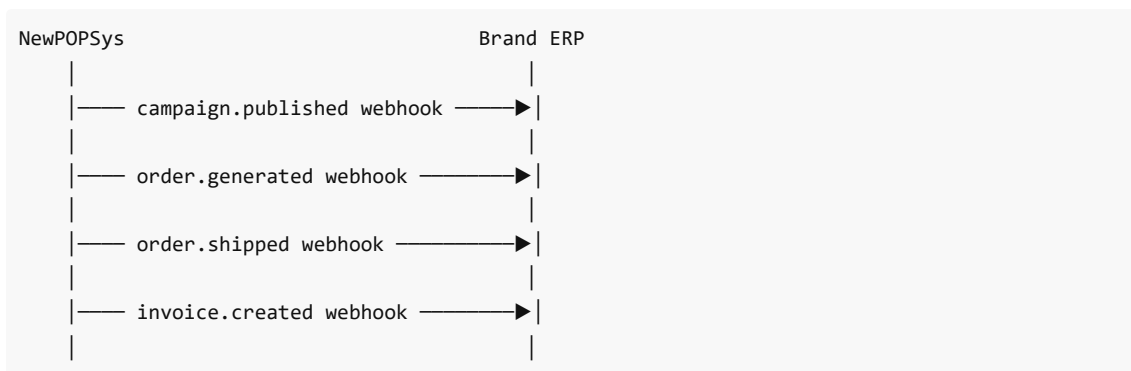
4. Brand ERP Integration

4.1 Supported ERP Systems

ERP System	Integration Method	Authentication
SAP S/4HANA	REST API / OData	OAuth 2.0 + Certificate
Oracle NetSuite	SuiteTalk REST	Token-Based Auth
Microsoft Dynamics 365	Web API	Azure AD OAuth 2.0
Sage Intacct	REST API	Session-Based

4.2 Integration Patterns

NewPOPSys sends event notifications to Brand ERP systems via webhooks. ERPs do not call back into NewPOPSys directly.



4.3 ERP Data Mapping

4.3.1 Campaign to ERP Project

NewPOPSys Field	SAP Field	NetSuite Field	Dynamics Field
campaignId	PROJECT_ID	entityId	msdyn_projectid
campaignName	PROJECT_DESC	name	msdyn_subject
budget	BUDGET_AMOUNT	projectedBudget	msdyn_plannedbudget
startDate	START_DATE	startDate	msdyn_scheduledstart
endDate	END_DATE	endDate	msdyn_scheduledend
brandId	CUSTOMER_ID	customer	customerid

4.3.2 Order to ERP Sales Order

NewPOPSys Field	SAP Field	NetSuite Field	Dynamics Field
orderId	SALES_ORDER_NO	tranId	salesorderid
orderDate	ORDER_DATE	tranDate	createdon
totalAmount	NET_VALUE	total	totalamount
shippingAddress	SHIP_TO_PARTY	shipAddress	shipto_composite
items[].sku	MATERIAL_NO	item	productid
items[].quantity	ORDER_QTY	quantity	quantity

4.4 Webhook Payload for ERP

```
{
  "event": "order.shipped",
  "timestamp": "2025-01-15T09:00:00Z",
  "data": {
    "orderId": "ORD-2025-001234",
    "externalOrderId": "SO-12345",
    "campaignId": "CAMP-2025-Q1",
```

```
"shippedAt": "2025-01-15T09:00:00Z",
"carrier": "UPS",
"trackingNumber": "1Z999AA10123456784",
"trackingUrl": "https://ups.com/track?num=1Z999AA10123456784",
"shipmentDetails": {
  "packageCount": 1,
  "totalWeight": {
    "value": 5.5,
    "unit": "LB"
  },
},
"estimatedDelivery": "2025-01-18"
},
"financials": {
  "shippingCost": 15.99,
  "currency": "USD"
}
},
"tenant": {
  "tenantId": "TNT-001",
  "brandName": "Acme Retail"
}
}
```

5. Identity & SSO Integration

5.1 Supported Identity Providers

Provider	Protocol	Features
Azure AD	SAML 2.0 / OIDC	Enterprise SSO, MFA, Groups
Okta	SAML 2.0 / OIDC	Universal Directory, MFA
Auth0	OIDC	Social Login, Custom DB
Google Workspace	OIDC	G Suite Integration
PingFederate	SAML 2.0	Enterprise Federation

5.2 SAML 2.0 Configuration

5.2.1 Service Provider Metadata

Attribute	Value
Entity ID	https://api.newpopsys.com/saml/metadata
ACS URL	https://api.newpopsys.com/saml/callback
SLO URL	https://api.newpopsys.com/saml/logout
Name ID Format	emailAddress

Signature Algorithm	RSA-SHA256
---------------------	------------

5.2.2 Required SAML Assertions

Attribute	Description	Required
email	User email address	Yes
firstName	User first name	Yes
lastName	User last name	Yes
groups	Group memberships	No
employeeId	Employee identifier	No
department	Department name	No

5.2.3 Group-to-Role Mapping

IdP Group	NewPOPSys Role	Permissions
newpopsys-admins	admin	Full access
newpopsys-brand-managers	brand_manager	Brand management
newpopsys-field-reps	field_rep	Mobile app, surveys
newpopsys-viewers	viewer	Read-only access

5.3 OIDC Configuration

5.3.1 OpenID Connect Settings

Setting	Value
Client ID	Provided per tenant
Redirect URI	https://api.newpopsys.com/oauth/callback
Response Type	code
Scopes	openid profile email groups
Token Endpoint Auth	client_secret_post

5.3.2 JWT Claims Mapping

```
{
  "sub": "user-unique-id",
  "email": "user@company.com",
  "name": "John Doe",
  "given_name": "John",
  "family_name": "Doe",
  "groups": ["newpopsys-brand-managers"],
```

```
"iss": "https://login.microsoftonline.com/{tenant-id}/v2.0",
"aud": "{client-id}",
"iat": 1705142400,
"exp": 1705146000
}
```

5.4 Session Management

Setting	Value
Session Duration	8 hours
Idle Timeout	30 minutes
Refresh Token Lifetime	24 hours
Concurrent Sessions	3 per user

6. Payment Integration

6.1 Supported Payment Processors

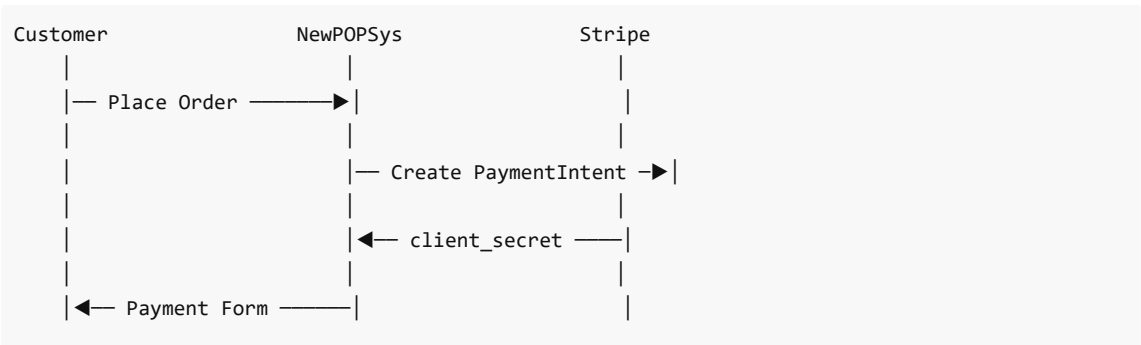
Processor	Use Case	Features
Stripe	Card Payments	PCI DSS, 3D Secure
PayPal	PayPal Payments	Express Checkout
ACH/Wire	B2B Payments	Net terms support

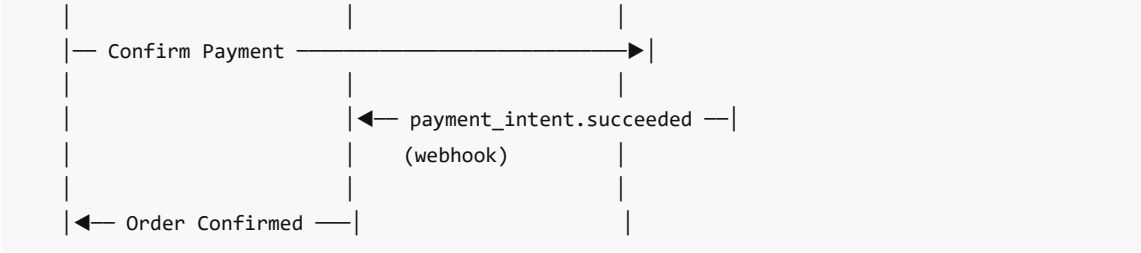
6.2 Stripe Integration

6.2.1 API Configuration

Setting	Value
API Version	2024-12-18.acacia
Webhook Version	2024-12-18.acacia
Payment Methods	card, us_bank_account

6.2.2 Payment Intent Flow





6.2.3 Webhook Events

Event	NewPOPSys Action
payment_intent.succeeded	Mark order as paid
payment_intent.payment_failed	Notify customer, retry
charge.refunded	Update order status
invoice.paid	Update subscription

7. Cloud Storage Integration

7.1 Supported Storage Providers

Provider	Use Case	Features
AWS S3	Primary storage	CDN, versioning
Azure Blob	Enterprise customers	AD integration
Google Cloud Storage	Alternative	Multi-region

7.2 AWS S3 Configuration

7.2.1 Bucket Structure

Bucket	Purpose	Lifecycle
newpopsys-artwork-{env}	Source artwork	2 years
newpopsys-photos-{env}	Survey photos	1 year
newpopsys-exports-{env}	Report exports	90 days
newpopsys-temp-{env}	Temporary uploads	24 hours

7.2.2 Pre-signed URL Generation

```
{
  "operation": "putObject",
  "bucket": "newpopsys-photos-prod",
  "key": "surveys/2025/01/RESP-001234/photo-001.jpg",
  "expiresIn": 3600,
  "conditions": {
```

```
"content-type": "image/jpeg",
"content-length-range": [1, 10485760]
}
}
```

7.2.3 CDN Configuration

Setting	Value
CDN Provider	AWS CloudFront
Origin	S3 bucket
TTL	24 hours (artwork), 1 hour (dynamic)
Signed URLs	Required for private assets

8. Notification Services

8.1 Email (SendGrid)

8.1.1 Configuration

Setting	Value
API Version	v3
Sender Domain	notifications.newpopsys.com
IP Warmup	Dedicated IP pool

8.1.2 Email Templates

Template	Trigger	Recipient
order-confirmation	Order created	Store manager
shipment-notification	Shipment created	Store manager
survey-reminder	Survey due	Field rep
issue-escalation	Issue unresolved 48h	Brand manager

8.2 SMS (Twilio)

8.2.1 Configuration

Setting	Value
API Version	2010-04-01
Sender Number	Short code or toll-free
Messaging Service	Enabled

8.2.2 SMS Use Cases

Use Case	Template
Delivery OTP	"Your delivery code is {code}. Valid for 30 minutes."
Survey Reminder	"Reminder: Complete installation survey for Store #{store_id}."
Urgent Issue	"URGENT: Issue #{issue_id} requires immediate attention."

9. Error Handling & Retry

9.1 External API Error Handling

Error Type	Retry Strategy	Max Attempts
5xx Server Error	Exponential backoff	5
429 Rate Limited	Respect Retry-After	3
401 Unauthorized	Refresh token, retry	1
4xx Client Error	No retry	0
Network Timeout	Immediate retry	3

9.2 Retry Schedule

Attempt 1: Immediate
Attempt 2: 1 second delay
Attempt 3: 5 seconds delay
Attempt 4: 30 seconds delay
Attempt 5: 5 minutes delay

9.3 Circuit Breaker Configuration

Setting	Value
Failure Threshold	5 failures in 60 seconds
Open Duration	30 seconds
Half-Open Requests	3
Success Threshold	2 consecutive successes

10. Security Requirements

10.1 API Credential Storage

Credential Type	Storage	Rotation
API Keys	AWS Secrets Manager	90 days

OAuth Tokens	Redis (encrypted)	Per session
Service Accounts	Vault	30 days
Certificates	ACM / Key Vault	1 year

10.2 Data Encryption

Data State	Encryption
In Transit	TLS 1.3
At Rest	AES-256
Secrets	AWS KMS / Azure Key Vault

10.3 Audit Logging

All external API calls are logged with:

- Timestamp
 - Target system
 - Operation type
 - Request ID
 - Response status
 - Latency
 - Error details (if any)
-

11. Monitoring & Alerting

11.1 Health Checks

Integration	Check Interval	Timeout
Shipping Carriers	5 minutes	10 seconds
ERP Systems	5 minutes	30 seconds
Identity Providers	1 minute	5 seconds
Payment Processors	1 minute	10 seconds

11.2 Alert Thresholds

Metric	Warning	Critical
API Error Rate	> 1%	> 5%
Response Latency	> 2 seconds	> 5 seconds
Circuit Breaker Open	N/A	Any open
Authentication Failures	> 10/hour	> 50/hour

12. Related Documents

Document ID	Title	Description
SRS-11.1	API Overview	Authentication, versioning, rate limiting
SRS-11.2	Internal APIs	RESTful endpoint specifications
SRS-11.4	Webhooks	Event-driven notifications
SRS-3.4	Integration Architecture	Technical integration patterns

Revision History

Version	Date	Author	Changes
1.0	2025-01-13	System	Initial version

11.4 Webhooks

Document Information

Field	Value
Document ID	SRS-11.4
Version	1.0
Last Updated	2026-01-01
Status	Draft
Parent Document	SRS-11 API Specifications

1. Introduction

1.1 Purpose

This document specifies the webhook system for NewPOPSys, enabling real-time event notifications to external systems including Brand ERPs, PSP MIS platforms, and custom integrations.

2. Event Types

2.1 Campaign Events

Event Type	Trigger	Payload
campaign.created	New campaign published	Campaign object
campaign.updated	Campaign details modified	Changed fields

campaign.activated	Campaign goes live	Campaign + stores
campaign.completed	All tasks finished	Campaign + summary
campaign.cancelled	Campaign cancelled	Campaign + reason

2.2 Task Events

Event Type	Trigger	Payload
task.assigned	Task assigned to store	Task + store
task.started	Store begins task	Task + timestamp
task.completed	Task marked complete	Task + evidence
task.verified	Task passes QA	Task + verification
task.rejected	Task fails QA	Task + rejection reason

2.3 Photo Events

Event Type	Trigger	Payload
photo.uploaded	New photo submitted	Photo metadata
photo.processed	AI analysis complete	Photo + scores
photo.approved	Photo approved	Photo + reviewer
photo.rejected	Photo rejected	Photo + reason

2.4 Order Events

Event Type	Trigger	Payload
order.created	Kit order generated	Order details
order.shipped	Order dispatched	Order + tracking
order.delivered	Delivery confirmed	Order + POD
shipment.exception	Delivery issue	Exception details

3. Payload Format

3.1 Standard Envelope

```
{
  "id": "evt_abc123def456",
  "type": "campaign.created",
  "api_version": "2026-01-01",
  "created_at": "2026-01-01T12:00:00Z",
```

```
"data": { "object": { } },
"metadata": { "brand_id": "brand_123", "tenant_id": "tenant_456" }
}
```

4. Security

4.1 Signature Verification

All webhooks signed using HMAC-SHA256. Header: `X-POPSys-Signature`

4.2 Timestamp Validation

Reject requests where timestamp differs from server time by > 5 minutes.

5. Delivery & Retry

Attempt	Delay
1	Immediate
2	5 minutes
3	30 minutes
4	2 hours
5	24 hours

After 5 failed attempts: webhook disabled, admin notified, events queued 7 days.

6. Configuration Endpoints

Method	Endpoint	Description
GET	<code>/api/v1/webhooks</code>	List webhooks
POST	<code>/api/v1/webhooks</code>	Create webhook
GET	<code>/api/v1/webhooks/{id}</code>	Get webhook
PATCH	<code>/api/v1/webhooks/{id}</code>	Update webhook
DELETE	<code>/api/v1/webhooks/{id}</code>	Delete webhook
POST	<code>/api/v1/webhooks/{id}/test</code>	Send test event

References

- [11.1 API Overview](#)
- [11.2 Internal APIs](#)
- [11.3 External Integrations](#)